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The biggest shed in the world: an approach to computer enhanced creativity

Abstract

When I was growing up we had a shed in the back yard, it was the place where any and everything we did not have an immediate use for was stored; there was no method to what was kept or where it was put. This created an environment where anything could be found with anything else, everything was familiar and nothing was sacred or precious: a true cornucopia from which much was made.
This paper looks at a role of the computer as an assistant to creativity through being a resource of information, opinions and interpretations and the conceptual spaces they inhabit, where the digitized object or idea is available to the artistic creative processes at a variety of levels and in a variety of ways.

Introduction

"Artistic expression forms an experiential bridge, crossing boundaries -- between people, communities, nations. The expression reifies the relationship between transmitter and receiver -- the relationship reifies the expression. (29 jan)" (author unidentified, http://www.acmc07.org/)

I made a cursory search for a definition of the verb *reify*, revealing two quite different interpretations, one from the 1974 Concise Oxford Dictionary(Fowler 1974), the other from Encarta World English Dictionary(Bloomsbury Publishing Plc. 1999), both common arbiters of word definitions in their respective times,

According to the Concise Oxford it means to "convert (person, abstract concept) into thing, materialize", and according to Encarta it means "to think of or treat something abstract as if it existed as a real and tangible object". Both dictionaries were and are well used in their different time frames, and both provide an interpretation of the respective world views of their times.

The distinction between these two interpretations resides in the phrases "to *think of* or *treat* something abstract *as if* … real", and to "convert into thing … [to] *materialize*" something; one is *conceptually* creative, the other *actually* creative. This demonstrates a change of meaning over time, the interpretation evolving appropriately to the technological and cultural developments within the societies using it. The Oxford definition indicates a time in which reality was less mutable and virtual realities were not considered as real possibilities, something was real or it wasn't and it was possible to *convert* the unreal into reality, perhaps through a creative act. The Encarta definition allows for a virtual construction, something can as easily be real as it can be *as if* real.

These two definitions, rooted in their times, indicate that definition(s), through which an understanding of the world is achieved, are dependent and mutable, and may be generated and used reactively and creatively when interpreting the world.[1]

The advent of the computer as a common tool was possibly the event that allowed the concept "*as If … real*" to become common, it is only recently that a virtual reality can be considered as a valid reality[2]. It is this tool that has redefined much of the processes used to interpret, influence, understand and be creative. For the musician the computer has had many influences, it provides ways to collate and organize sounds, ways to generate sounds and, most importantly here, ways to reorient practices.

Artistic creativity

The computer-oriented musician/composer as belongs to a subset of the larger set 'artist'. It can be assumed that the 'artist' is essentially a communicator, who sees publicly expressing their ideas as implicit in their role. Leo Tolstoy saw that "Art begins when one person, with the object of joining another or others to himself in one and the same feeling, expresses that feeling by certain external indications." (Camp 1998)[3]. Of course the reasons for making art are as many as there are artists, and it is possible to substitute other words such as, perhaps 'concepts' or 'ideas', for "feeling", making his statement more suitable and accurate in some cases.

An artist creates something that may be historically new, in that the work, and possibly the concepts that allowed it, have not previously existed, and it may also be personally new, in that the artist has created something that they have not created before and based on concepts new to themselves. In the act of creation, there is often a need to redefine approaches and systems assumed as part of the creative act, and this is a significant part of the artist's role.

The Art(ist) as re-definer

When making something new, a *re*defining of what is known or believed to be known is often essential; the case above indicates a *re*defining of a word resulting from broad societal changes.

One of the functions of the artist is to be a re-definer, reinterpreting the world for themselves and subsequently for others through their work. For example: by painting a pipe and stating within the painting that it is not a pipe Magritte confronted the notion of representative painting; by renaming and relocation a urinal Duchamp created the 'readymade', requiring the viewer to see objects

differently; and by having a musician play silence in a concert hall Cage caused the world to become musical for 4'33". By reinterpreting and redefining traditions in their respective practices these artists questioned many traditionally held beliefs and understandings of art and creativity. In each of these cases the creative works caused an iconoclasm of some sort, forcing a broader reinterpreting and redefining of viewpoints within the various arts practices and perhaps further.

Within the creative process, the artist also often reinterprets and redefines themselves, their relationship to their work, and to their world. This can occur as the artists discovers new paradigms through self reflection of and through examining their own work. It may also occur as new technologies, and the new conceptual frames that they invoke, become available.

The ubiquity of computer-based technology in technologically developed societies offers the artist residing within those societies opportunities unthinkable a generation ago. Being able to access intellectual, conceptual, physical materials from other societies offers a plethora of opportunities for redefining and reinterpreting, either of those materials or through those materials. It also offers opportunities to generate wholly new conceptual frames and the subsequent works that may either be developed from within or as a result of those frames. This is one of the computers greatest contributions to the creative artist.

The Artist as creative agent

Margaret Boden sees creativity as fitting two possible areas, H(istorical)-creativity, "creating a concept which has never been created before at all [and P(sychological)-creativity], a concept which has never been created before by a specific creator." (Boden 1995; Wiggins 2001). Both of these areas are of interest to the computer-based artist, in that they are working with technology where H-creativity is occurring at a meta level and P-creativity is occurring a more local/personal level. Of course, there is considerable overlap between these two modes, they are often looping to equally influence conceptual and technological frames coincidentally.

In Boden's terms a more creative result is one that transforms a conceptual space, not one that just the explores or exploits a conceptual space, however expert or novel that exploration or exploration may be. According to her "dropping a constraint is a general heuristic for transforming conceptual spaces." (Boden 1995) An example she gives of this is Friedrich August Kekulé redefining his view of the compound benzene in order to better explain it[4] (Boden 1995); perhaps it is finding the right constraint to drop that creative genius resides.

The creativity of people such as Kekulé, Albert Einstein or John Nash lay in their ability to reinterpret and then redefine; having impact by generating a change in conceptual spaces that for the most part did not have immediate tangible outcomes. However, what these changes allowed for was the exploitation and reification of and within the newly created conceptual spaces, providing for and resulting in new understandings and tangible outcomes. Artistic creativity requires a tangible outcome; it requires that something unreal be first be formulated and considered as possibly real by the artist, who then makes it real.

The expression of the creative artist is through being actually creative, the reification, in the 1974 Oxford sense, of the artistic impulse. By bringing abstract things into a state that can be witnessed, the artist shares the impulses that caused the creative act, making their creativity tangible[5].

When embarking on the creative process the artist may have a notion that they think is worth sharing with others, and the reasons for this desire are manifold. On another hand, the artist may be experimenting within their expertise such as Jackson Pollock's development of 'drip' painting into his technique, and through doing so develop or discover something that they believe is worth sharing Or the artist may be endeavoring to broaden their expertise in new or allied fields and in doing so generate works that are novel to themselves and to the audience of that field, composer/choreographer/installation artist Meredith Monk is an example here.

The audience as creative agent

It is also very much the case that an artist's, or otherwise creative person's, work may not be fully appreciated or understood in their times. Vincent van Gogh is an example of an artist whose work was not appreciated in his lifetime, selling one or two paintings before his death. Other artists such as Anton Webern or Raymond Scott, while perhaps slightly more successful in their lifetimes than Van Gogh, still were not recognized for the contributions they made to their respective fields during their lifetimes.

Another example of creativity not being recognized is Heron of Alexandria's use of steam power somewhere between 150 BC and 250 AD. The majority of his work with hydraulics was used in the development of entertainments, creating fountains, raising platforms, magically opening doors, and so on. Perhaps the most potent, but unrecognized, of his designs was the

> "aeolipile … a hollow sphere mounted so that it could turn on a pair of hollow tubes that provided steam to the sphere from a cauldron. The steam escaped from the sphere from one or more bent tubes projecting from its equator, causing the sphere to revolve. The aeolipile is the first known device to transform steam into rotary motion." (O'Connor and Robertson 1999)

While the aeolipile was seen as a toy at the time of its invention it is now easy to see that it is actually a steam-powered engine; made useful by simply attaching an axle to the revolving sphere.

That Heron and his contemporaries did not see this possibility shows, perhaps, a lack of understanding of the conceptual frameworks required to make this connection.

In 1690 Isaac Newton explained the Laws of Thermodynamics, the third, "Every action produces a reaction equal in force and opposite in direction", having particular relevance to Heron's aeolipile, perhaps providing the required conceptual frames for its exploitation. It was shortly after this understanding of nature's behavior that Thomas Savery was able to exploit Heron's toy in creating the steam engine(O'Connor and Robertson 1999; Lahanas 2004; Forrester 2005).

These are examples of the contemporary viewer not valuing a creative work simply because they do not have the tools through which to access and assess the work. Unfortunately, and possibly most profoundly, the predominantly market driven society all audiences exist in defines the immediate value of any creative work, artistic or not, and to a large extent, how the work should be perceived.

It is also possible that the creative act can reside within the viewer, making them as much a part of the creative process as an author. This collaboration in the creative process can be expressed through a variety of interpretations; photography is an example where the often viewer creates the artwork. In this case the viewer captures and interprets on film an existing object at a 'decisive moment' in such a way as to make it an artwork, the works of Henri Cartier-Bresson, who has exceptional skill in capturing and interpreting the 'right', unedited, "readymade", moment, such as *Rue Mouffetard, Paris, 1954*, are an example. (Mraz 2003)This is a case where the viewer created an artwork simply through observing and capturing an unrehearsed or un-devised event.

The more traditional audience plays a more passive but effective role in creating an artwork simply through interpreting the artwork from their own perspective(s). This means that the artist loses their position as soul author and arbiter of their work through the very act of expressing and sharing their creative impulse(s).

An example of this process is my (re)interpretation of Cage's *Imaginary Landscape No. 4*. The piece as performed now would vary significantly the premier performance in New York in 1951. "When one listens to [or performs] the work, it is obvious that one cannot predict what will be heard, which is exactly what Cage was aiming at with this composition. [my underline]"(Unknown 2006) Obviously the content would be vastly different in a contemporary performance from what Cage could possibly have imagined when composing it; but it is also possible that a contemporary interpretation of the work will be significantly different from what he might have assumed or predicted.

My understanding of the work shifted dramatically while driving from Phoenix to San Diego in 1993. On that trip I set the car radio to scan and it took a long time to get from one end of the dial to the other, spending about ten seconds on each station. This led me to reinterpret Cage's work to be about social, temporal and locational change, where a four-minute performance would give a very precise indicator of each of those three elements. In this way I, as audience, captured, redefined and reinterpreted another artists work, personalizing it and claiming it as my own.[6]

Computer creativity

The computer forms a robust fulcrum around which conceptual spaces can spin or teeter. As an artist the computer allows an interrogation and exploitation of very many conceptual spaces, and the myriad offshoots that they can engender, with comparably less effort than was required in pre-computer times. This is potentially one of the most valuable attributes of the computer to the creative person. It offers similar meta-explorations to that undertaken in the field conceptual art, which

> "sought to analyze the ideas underlying the creation and reception of art, rather than to elaborate another stylistic convention in the historical succession of modernist avant-garde movements. Investigations by conceptual artists into networks of signification and structures of knowledge (which enable art to have meaning) … examine the interstice between visual and verbal [or sonic] languages as semiotic systems." (Shanken 2002)

This exploration creates an area where creativity itself is being explored and forms a framework in which researchers, such as Boden and Minsky (Minsky 1982), have made inquires. There is a sense that the pathway to developing effective AI creativity is through gaining a clear understanding of human creativity and modeling it. This approach has broadened over years, the call for papers for the MUSIC-AI 2007 International Workshop on Artificial Intelligence and Music gives a list of methods used of AI approaches to musical creativity including:

> "cognitive modeling, data mining and classification, expert systems, generative systems, grammars, fuzzy logic, genetic algorithms, hidden Markov models, intelligent agents, inductive logic programming, knowledge representation, knowledge-based systems, machine learning, neural networks, constraint satisfaction, and planning".(Unknown 2006)

This list indicates the breadth of techniques used and available to get a creative outcome from a computer. In exploring each of these approaches, an understanding of "networks of signification and structures of knowledge (which enable art to have meaning)" may be gained.

The Computer as re-definer

The computer as re-definer is not really that different from the artist as re-definer. The home movie maker filtering their movies, relocating the images and removing anything unwanted is an example of redefining and reinterpreting the original material made easily available through computer technology.

This kind of rebuilding is prevalent in sound, for example: sounds can have the place in which they were captured redefined as another place, real or not; they can be splintered and rebuilt into completely different sounds that would be unrecognisable when compared to the original. Sounds can be turned into visuals and visuals into sounds; movement can be re-defined as any of the above. In these examples the artist is the generator and cause of the changes.

The computer as creative agent

The computer can assist the artist in the creative process and in doing so generate outcomes that are personally new and potentially historically new in many ways. These include, perhaps most potently but often less recognized, the extraordinary access to and rapid transference of information afforded by computer technology. It is now possible for an artist to quickly develop, or become part of, a tightly focused yet large community, in which ideas are quickly discussed, interrogated and evaluated. Up until the advent of computers, and subsequently the Internet, this kind of intercourse was difficult and time consuming.

Use of computers as an assistant to the creative process, or as a creative machine themselves can be seen in the work of programmers, scientists and engineers such as Steven Thaler, whose "Creativity Engines are neural network systems that display creative behaviour; this effect is achieved by introducing a precise degree of random perturbation into the neural networks"(DeGracia 1999). This is a process that attempts to replicate an aspect of the creative process; the jumbling of data within set constraints and allowing new logics to appear. (Thaler 1996; Hesman 2004)

It is also possible to use the computer as a tool that reconstitutes the work of certain composers into new compositions reminiscent of those composers. (Cope 1999; Cope 2003; Cope 1992), or that generates entirely new compositions via algorithms created by composer.

There are many programs that will generate what may be called creative artworks such as Harold Cohen's painting program AARON and Ray Kurzweil's poetry generating program (Kurzweil 2000; Cohen 2001; Kurzweil 2001); and the music generating programs of very many computer oriented composers[7]. Bruce Buchanan provides a list of creative programs that expands those above to include the areas historical science, chemistry and mathematics. (Buchanan 2001)

One area of creative artistic activity that claims, with some justification, to have integrated computer technologies into the entire creative process is dance, Johannes Birringer claims that

> "dance has taken the lead, among the theatrical arts, in absorbing technology as a creative tool, affording dancers and technologists the opportunity to explore interactive environments, virtual places, and integrated methods that have shifted artistic process." (Birringer 2002)

Here the technology allows for an, occasionally seamless, integration between the modalities of movement and sound requiring the composer create outside of traditional musical paradigms.

Problems with computer creativity

The perpetual nascence of computer technology makes it difficult for the computer based art form to reach a maturity of expression similar to that of other art form, oil painting for example. Not being static long enough to develop a maturity similar to that of more well established practises may simultaneously be the computer's most valuable and frustrating attribute.

There is also, among some computer musicians, an infatuation and focus with the technology, this can be quite appropriate as the technology in some ways bounds and expands the possibilities of the practice. However, there is an infatuation with the next and newest does not allow development of expertise similar to that of an instrumentalist, and that can occasionally mask or subvert the reason for the art making. An extreme metaphor would be architects focussing on building equipment as an integral part of their artistic discourse.

My work

Three recent works, *5*, *20* and *41*, signify developments in my own art making that result from new interpretations of developments in computer technology. The initial notions that caused the works related to the supposed 'impersonalness' of digital technology, particularly when used as a tool in art making. It is easy to see that the digital artist must have a foot in two camps, that of the 'artist' who invents the cause of the work, and of the 'technician', who implements the work. These positions are mutually informing and overlapping, causing a practise built on shifting sands. Personally, I find these shifting sands one of the most intriguing and fascinating aspects of computer based art making, creating a continued sense of displacement and wonder.

The area of artistic expression I have been looking into recently, and wanting to explore in these works, is the portrait. This is an area of art making well explored in visual arts but perhaps less so in the computer based arts. Two other areas of interest are the 'readymade', which is very much explored in the digital arts, evidenced by the prodigious use of sampling in music and freely available images in the visually oriented arts; and

finally glitch, deliberately creating situations where the computer behaves in an unpredictable way due to inabilities of the hardware and/or software.

When beginning the exploration I worked with images and sounds of my immediate material and virtual environments. I also looked for digital portraits on the Internet, choosing those available in Flickr (2007) under the creative commons license as being easy to access and having much variety.

What I found most interesting was the vast number of people making images of themselves and others available for use by anyone else[8]. My predisposition regarding photographs I have of friends and family is that they are viewed in highly controlled circumstances to invited viewers; I found the willingness to make what I considered intimate photos, such as wedding or photos of the children strangely disquieting and intriguing, automatically thinking that these images were for a select audience. The other thing that I found as intriguing was the number of posed 'art' portraits available; people were obviously being creative with the medium and broadcasting their creative efforts widely.

Over January 2007 I gathered a number of images of faces from Flickr and began to adjust them with the Max/MSP/Jitter (Zicarelli 2007) program using various processes, finally centering on the magnifying object "jit.rota" and reducing images to areas around one hundred thousands of their size. This resulted in significant, and unpredictable, alterations in the images. Occasionally the images would recognizable but appear granulated or pixilated in various ways. The unpredictability of the outcomes was, I am assuming, due to the interactions of the hardware and software of the computer. In order to test this I ran the same patcher on two very different computers, a 2GHz Intel Core 2 Duo iMac with 2 GB of RAM running OS 10.4.7 and Max 4.6.2 and Jitter 1.6.2, and an 867 MHz G4 PowerPC laptop with 640 MG of RAM running OS 10.4.8 and Max 4.5.7 and Jitter 1.2.4. The program worked on both machines but the outcomes, while quite similar, showed some differences. The most profound differences were when the works were rendered to DVD.

These differences were mainly in the visual frames being presented in performance and when being recorded. Importantly these differences did not significantly affect the ideas that formed the artistic intentions.

Conclusion

Using computers in the creative process can follow many paths, each as idiosyncratic, individual and personal, as there are composers. By using methods such as those discussed in the MUSIC-AI 2007 call the composer can be introduced to new ideas, approaches, musical paradigms and so on. However there is as much to be gained by using broader, less esoteric, computer technology to inspire these things within the individual creator.

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Endnotes

[1] In fact, the word does not exist in the 2003 Oxford Australian Integrated School Dictionary, used by current secondary school students.

[2] People now spend real money for virtual clothes in *as if* realities such as 'Second Life'

[3] Tolstoy had many other comments on "Art" that are succinctly listed by Camp.

[4] Boden recounts that Kekulé came across his method of understanding in a half waking dream, where apparently unrelated images caused the required insight.

[5] By tangible I mean something that can be witnessed, and include works that have an ephemeral existence, such as a dance or musical performance or an installation.

[6] Without negating Cage's authorship.

[7] Obviously, this list would be enormous were I to include all of the computer-based artists who use the tool in their creative works and explorations.

[8] It must be noted here that the open creative commons license is the default when posting images on Flickr.